KEERTHISAMEERA IMMANNI

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EDUCATION:

Northeastern University, Boston, MA

Master of Science in Telecommunications Systems Management **May 2017**

Coursework: Data Networking, Linux for Networking Engineers, Fundamentals of Computer Networking, Telecommunication Architecture

Jawaharlal Nehru Technological University, Tuni, India

Bachelor of Technology in Electronics and Communication Engineering **May 2014**

Coursework: Telecommunication Switching Systems, Digital Communications, Cellular and Mobile Communications, Computer Networks

TECHNICAL SKILLS:

Networking Protocols	: DNS, HTTP, DHCP, SSH, FTP, TCP/IP, UDP, RIP, OSPF, EIGRP, HSRP, STP,RSTP,ICMP, ARP, PPP.
Technologies	: VLAN, Ethernet, VLSM, VPN, Frame Relay, IEEE 802.11
Languages	: C, Python, Shell Scripting (Bash)
Operating Systems	: Windows, MAC, Linux (Ubuntu, Red Hat)
Tools	: Cisco Packet Tracer, Wireshark, VMWare Workstation, Putty

CERTIFICATIONS:

- Cisco Certified Network Associate in Routing and Switching (200-125) - CSCO13080501 **Valid till February 2020**
- Certification on "Interactive programming in Python" from Rice University

WORK EXPERIENCE:

Intern (Bharat Sanchar Nigam Limited, Vijayawada, India) **August 2014 – February 2015**

- Analyzed and administered the working of PSTN switching system by visiting various telephone exchanges/Mobile towers.
- Learned theoretical as well as pragmatic concepts on working of telephone exchanges and different technologies like GSM, CDMA, Broadband, 2G and 3G and also Data communication technologies like TCP/IP, MPLS.
- Documented reports on GSM Planning, Installation and operation of BTC, BSC and MSC.

ACADEMIC PROJECTS:

Roll Your Own Content Delivery Network **December 2016**

- Deployed own CDN and implemented a DNS redirection technique to send clients to the replica server with the fastest response time.
- Implemented a HTTP server in Python that hosts on Amazon EC2 servers all over the world as replica servers to return the content requested by clients, either from its cache or by fetching it from the origin.
- Developed a mapping system that uses geo-location and cached data at servers to determine the best replica server for each client request.

Performance Analysis of TCP Variants **November 2016**

- Analyzed performance of TCP variants Tahoe, Reno, New Reno and Vegas in NS-2 by varying test conditions and also fairness of a TCP variant against another variant with constant UDP flow.
- Calculated and documented performance parameters such as throughput, drop rate and latency by evaluating trace files from NS-2 simulation and plotted graphs for all TCP variants.

Web Crawler in Python **October 2016**

- Deployed a Web Crawler script for sending HTTP requests to a website through sockets using python.
- Found secret flags by crawling through relevant pages using BFS algorithm and also avoided loops using respective variables.

Simple Client TCP socket programming in Python **September 2016**

- Developed a script for the client machine to communicate with server using TCP connection
- Implemented SSL communication and analyzed packets using Wireshark for troubleshooting purposes.

Linux Bash Shell Scripting | Network Configuration Automation **April 2016**

- Developed a Configuration script to automate the process of configuring interface, hostname, DNS and sysctl parameters using awk, sed, grep, regexp commands by handling all invalid entries & error checks and logged the new entry updates to a file for debugging purpose.

Ad-Hoc Network for DNS, DHCP, Firewall and Webserver using Linux **December 2015**

- Implemented DHCP server using "isc-dhcp-server" for IPV4 and IPV6 and add-ons like VPN, Mail Server and NTP in the network.
- Configured backup and webserver using apache2 for HTML pages and designed firewall using IP tables for blocking traffic.
- Using bind9 configured Master and Slave DNS server and performed zone transfers like AXFR and IXFR in a team of four.

Implementation of Network Design for an Organization using Cisco Packet Tracer **November 2015**

- Designed optimized network for an organization and its branches in several locations by using IP address with CIDR and IP Subnetting.
- Designed VLANS and switch redundancy using MSTP also implemented HSRP, OSPF, Port security and Frame relay and ACLs.